When the Gradient Sees Rank: Provable Necessity, Causal Recruitment, and Exact Composition in Trained Matrix Memories

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Abstract

Matrix-valued fast-weight memories make rank a structural budget: the directions a state spans bound what it can bind and compose. A recent negative result reported that a matrix latent bolted onto a vector-pretrained reasoner never uses its rank, but the probed task admits a rank-1 solution, leaving the general question open. We answer it on testbeds where the exact solution provably requires $\mathrm{rank}(Z) \geq K$ and where every rank-evading shortcut is closed by construction: the readout is pinned to exact continuous recovery, since argmax decoding lets a rank-m memory store roughly md associations; a single-matrix-state bottleneck forecloses position decomposition; and no cell is called dead without re-testing at 2–2.5× budget. Under these conditions gradient descent recruits the rank the task demands in this architecture family: learned effective rank tracks K across the tested grid (Spearman $\rho = 1.0$ at $d = 16$), train-time rank caps show a causal step at $k \approx K$ (at $d = 8$, $K = 4$: rank 3 gives at most 0.0004 recovery, rank 4 gives 0.97), and the trained operator composes exactly through 21-fold self-application in four of five seeds. An entity-subspace decomposition supplies the mechanism, restricted rank exactly K on the key subspace, and in the single converged rank-starved seed, the depth-decay curve is predicted from its eigenspectrum alone.

Keywords: effective rank, fast weights, associative memory, composition

1. Introduction

Matrix-valued latents, in continuous chain-of-thought models (Hao et al., 2024) and fast-weight architectures broadly (Schlag et al., 2021), carry a structural observable a vector does not have: the *rank* of the state, which should track how many items the state holds and, when constrained below what the task needs, destroy performance. A recent negative result tested this on a matrix-CODI model (a $d \times d$ latent bolted onto a vector-pretrained GPT-2) and found rank-k truncation curves flat across four training regimes and four readouts (Larson, 2026); its own stated confound is that ProsQA admits a rank-1 solution, so flatness says nothing about tasks that provably require rank. We close the confound: matrix-native models trained from scratch on tasks whose exact solutions carry a provable $\mathrm{rank}(Z) \geq K$ lower bound, under a readout and architecture built so nothing short of genuine rank recruitment can pass. The contributions: the *teeth* that keep the bound a necessity result (§2); recruitment of rank tracking K and its causal necessity under train-time caps (§3); exact composition to depth 21, its invariant-subspace mechanism, and the depth-amplification signature (§4); and an exactness frontier in d with a budget-extension rule (§5).

2. The Bound and the Teeth That Keep It a Bound

Task and bound. Each episode presents K key→value bindings, then queries one; keys and values are fresh per episode (Gaussian, normalized, or exactly orthonormal following Mezzadri, 2007), so evaluation uses never-seen bindings. Exact recovery $Zk_j = v_j$ for K bindings with linearly independent keys and values forces $ZK_{\text{mat}} = V_{\text{mat}}$, hence the classical bound $K = \text{rank}(V_{\text{mat}}) = \text{rank}(ZK_{\text{mat}}) \leq \text{rank}(Z)$ (Kohonen, 1972; Anderson, 1972). We measure whether a trained, bottlenecked model *discovers* this rank under plain SGD, and whether it is causally load-bearing.

Tooth 1: exact continuous recovery, never argmax. Nichani et al. (2025) construct a rank-m linear memory storing approximately md associations under interference-tolerant argmax decoding; under that rule a rank-1 state passes at every K we test and the bound collapses. We therefore pin the decoder to the linear unbind itself, pred $= Z \cdot k_j$, scored by an absolute bar (rec@0.9: the fraction of queries with cos > 0.9), never an argmax and never a learned readout that could launder rank.

Tooth 2: a single-state bottleneck, verified behaviorally. "Hold K items" is satisfiable in full attention by attending to K raw-input positions at rank 1 each, so all information funnels through one $d \times d$ state and the decoder is a pure function of (Z, query): after the encoder writes Z, the raw-input cache is corrupted and the decode is confirmed bit-for-bit unchanged (a gradient blank-out test, not a mask-shape check).

Model and controls. A Transformer encoder over the binding tokens, with d learned row-reader latents producing the rows of $Z \in \mathbb{R}^{d \times d}$: permutation-invariant, able to express any rank up to d, and not hard-coding $\sum_j v_j k_j^\top$. Every model is under 1M parameters (the $d = 16$ composition model: 170,896). Effective rank (entropy of the normalized singular-value spectrum) is the pre-registered primary metric; the primary causal control constrains Z to rank k at *every training step* (spectral projection), not post hoc.

3. Rank Tracks K and Is Causally Necessary

Recruitment. Without any rank term in the loss, the trained d=16 state's effective rank rises one-for-one with binding count across a ten-point grid from K=1 to K=16 (Spearman $\rho = 1.0$; 2.42 at K=1, 8.20 at K=8, 15.09 at K=16), leaving the pre-registered $[0.7K, 1.3K]$ band only at K=1, 2, by overshoot; beyond $K = d$ rank saturates and declines. A subsequent, larger replication sweep is directionally identical.

Causal necessity. Training with Z constrained to rank k throughout (Figure 1): at d=8, K=4 the step is exact, $k \leq 3$ yields rec@0.9 ≤ 0.0004 and $k = 4$ yields 0.97, a discontinuity at the provable bound. The d=16 cells show the same knee near $k \approx K$ with mild post-knee non-monotonicity from seed noise; the replication resolves the K=12 transition into a ragged ramp, not a clean step. The two halves answer Larson (2026): the earlier rank-blindness was a property of a rank-1-solvable task, not of the gradient.

4. Exact Composition and the Invariant-Subspace Mechanism

Binding is a single lookup; the composition variant relates the K entities by one Hamiltonian K-cycle π, so a hop-depth-h query means recovering $v_{\pi^h(i)}$ by h-fold self-application of the same trained operator, $Z^h k_i$, with no learned per-hop parameters. Training sees hops

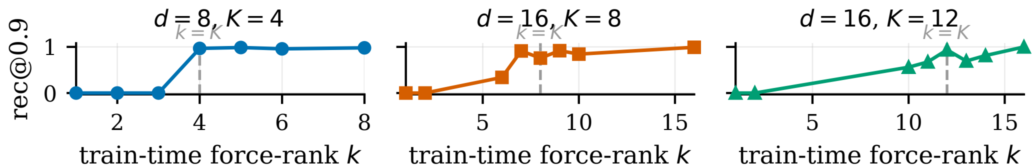

Figure 1: The primary, pre-registered causal test: rec@0.9 under train-time force-rank k, regenerated from the archived sweep (dashed line at $k = K$). The step lands at $k \approx K$ in every cell whose unconstrained model reaches exact recovery.

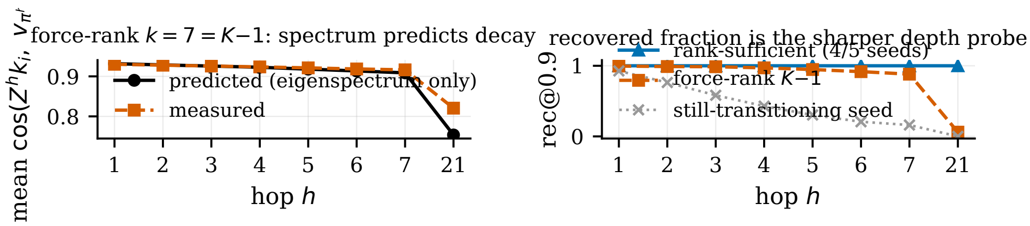

Figure 2: Depth as a spectral-exactness amplifier (d=16, K=8). *Left:* mean cosine for the converged force-rank k=7=$K-1$ seed against the eigenspectrum-only prediction (no raw keys). *Right:* rec@0.9 by hop for the four converged rank-sufficient seeds (flat at 1.0), the rank-starved operator, and the still-transitioning seed.

$\{1, 2, 3\}$; evaluation adds held-out $\{4, 5, 6\}$ and probes $\{7, 21\}$ ($d = 16$, $K = 8$, orthonormal keys). A single full cycle is load-bearing: under a general random permutation, nominally held-out depths collapse into trivial or in-distribution queries modulo its short cycle lengths (Appendix A).

Exact composition. 4/5 unconstrained seeds reach rec@0.9 $\geq$ 0.9996 at every tested hop (3/5 at 1.000 throughout); the fifth is still transitioning at budget, decaying from 0.93 (h=1) to 0.16 (h=7) and 0.0001 (h=21).

Depth amplification. At force-rank $k = K - 1$ (provably insufficient), the one converged seed passes shallow hops at mean cosine 0.92–0.93 ($h \leq 5$), above the 0.9 bar, and collapses only under depth: rec@0.9 falls 0.996 $\to$ 0.881 $\to$ 0.060 at $h = 1/7/21$. The mechanism is spectral and predictable: from the operator's eigenspectrum alone, predicted cosine matches the measured curve within 0.008 through $h = 7$ (Table 1).

The invariant-subspace mechanism. Whole-matrix effective rank of the converged solutions reads the full ambient dimension (16.0 of d=16; stable rank 14.6–15.6); the whole-matrix number is the wrong instrument. Block-decomposing the trained Z against the ideal cycle's K-dimensional entity subspace $\mathcal{E}$: the restricted operator has effective rank 7.999–8.000 (maximum 8), matching the ideal scaled cycle to a scale-corrected residual of 0.7–2.4%; leakage out of $\mathcal{E}$ is below 1.5% of the state norm; the complement block is full-rank and non-contractive yet dynamically invisible: a query starting in $\mathcal{E}$ never leaves it

(Appendix B). Restricted to the subspace the task uses, "rank tracks K" holds exactly; dead force-rank seeds fail the same restricted test, so their collapse is real.

5. The Exactness Frontier, and When to Trust a Dead Cell

Budget reversals. Three independent axes each declared a cell dead at its first step budget before an extension reversed the verdict: composition at $K{=}8$ (1/5 seeds converged at 20K steps; 4/5 at 40K), composition at $K \in \{12, 16\}$ (0/5 at 40K; 3/3 and 2/3 fresh seeds at 80K), and d-scaling (an 8K-budget wall at $d \geq 32$ dissolving at 20K, onset 6–16K). The rule, *a cell is not dead until re-tested at 2–2.5× budget*, is necessary but not sufficient: $K{=}16$'s remaining stuck seed stayed dead through 120K steps, and the $d{=}32$ cells below plateau flat yet still fail the pre-registered bar.

The frontier that survives is not a trainability wall but an *exactness* frontier. Fixing $K = d/4$ at encoder width $h = 64$ (every cell confirmed flat before being read as converged), mean recovery cosine falls monotonically across d: 0.877/0.909/0.915 at $d{=}32$ (three seeds, 100K steps), 0.7196 at $d{=}48$ (100K), and 0.3882 at $d{=}64$ (150K); the exact-match metric is harsher, with rec@0.9 0.413/0.632/0.653 at $d{=}32$ (all below the pre-registered 0.7 pass bar), 0.002 at $d{=}48$, and 0.0 at $d{=}64$. Rank recruitment and exactness separate: at $d{=}32$ recruited rank reaches 91–97% of target while exact recovery stalls below the bar.

6. Related Work, Limitations, and Outlook

Nichani et al. (2025) is the nearest neighbor: a hand-built existence construction under interference-tolerant argmax decoding; this work pins the readout so that construction cannot pass, then measures and intervenes on the rank SGD recruits (Barnfield et al., 2026 give static-memory thresholds). Nazari and Rusch (2026) and Sun et al. (2026) measure effective-rank dynamics in pretrained linear-attention models observationally (an *upper* bound $\mathrm{rank}(S_t) \leq t$); here the write count is controlled and necessity is causal. Mishra et al. (2026) train matrix-state RNNs on S_3 with no rank measurement or intervention (their own vector-state control matches); Grazzi et al. (2025) and Siems et al. (2025) (the Householder extension of Yang et al., 2024) characterize the per-token *transition* operator, not the trained *state's* own rank, the quantity here. Continuous-CoT capacity results are existence constructions, not measurements (Zhu et al., 2025; Gozeten et al., 2025); the compositional-generalization caution (Dziri et al., 2023; Wang et al., 2024) is met with an exact eigenstructure match.

Limitations. Every experiment is synthetic and under 1M parameters; generality at scale, and across state-writing architectures (recurrent fast-weight, attention-readout, associative-memory), is untested. The depth-amplification contrast rests on the single converged force-rank-($K{-}1$) seed; the other two at that cell collapsed under a documented numerical instability in the spectral-projection backward pass, so it is a case study consistent with the spectral prediction, not an $n{>}1$ mechanism; the $d \geq 48$ frontier cells rest on single seeds. The registered falsifiers stand (Appendix C). A concurrent companion submission (group-composition state tracking) shares no figures or tables but restates this paper's recruitment, causal-necessity, and composition numbers as background for its distinct contribution, a group-dimension rank law this paper does not make.

Appendix A. Depth 21 Under the Single K-Cycle

Under a single 8-cycle, $\pi^{21} = \pi^{21 \bmod 8} = \pi^5$, so the nominal depth-21 probe shares its *target* with the already-tested depth 5; the two are not distinct group-theoretic queries. What depth 21 measures is numerical stability under 21 sequential self-applications of the trained operator. In the four converged seeds rec@0.9 reads ≥ 0.9996 at both h=5 and h=21, so repeated self-application does not amplify their residual error; in the rank-starved operator the same 16 additional applications compound a passing shallow-hop cosine into collapse (0.947 at h=5 to 0.060 at h=21 in recovered fraction), which is exactly the amplification claim of §4, not a generalization claim to unseen permutation targets.

h	predicted cos	measured $\overline{\cos}$	measured rec@0.9
1	0.9317	0.9303	0.996
3	0.9258	0.9259	0.986
5	0.9181	0.9212	0.947
7	0.9089	0.9163	0.881
21	0.7536	0.8206	0.060

Table 1: The rank-starved ($k = 7 = K - 1$) operator's depth-decay curve: cosine predicted from the trained operator's own eigenspectrum (no raw keys) against the measured values, with the recovered fraction alongside. Every hop through $h = 7$ matches within 0.008; at $h = 21$ the per-item distribution widens and the exact-match metric collapses while mean cosine moves little.

Appendix B. Per-Seed Subspace Decomposition

seed	effrank(A)	residual	leakage (% of $\|Z\|$)	effrank(D)	$\rho(D)$
s1	8.000	0.7%	0.44%	8.000	1.38
s2	7.999	1.9%	1.06%	8.000	1.27
s3	7.999	1.5%	0.87%	8.000	2.86
s4	7.999	2.4%	1.45%	7.999	1.02

Table 2: Entity-subspace block decomposition of the four converged K=8 composition seeds (means over four evaluation episodes), recomputed from the archived Z-dumps. A is the operator restricted to the K-dimensional entity subspace; residual is the scale-corrected Frobenius distance to the ideal cycle after fitting the single isotropic scale that cosine scoring cannot see; leakage is the cross-block norm $\|C\|$ as a fraction of $\|Z\|$; D is the complement block, full-rank and non-contractive ($\rho(D) \geq 1$) yet never visited by a query trajectory that starts in the entity subspace.

Appendix C. Reproducibility

Training, evaluation, and analysis code is available at https://anonymous.4open.science/ (anonymized for review). Raw per-run JSON archives (dense training-step checkpoints, not summary rollups) back every number in this abstract; each figure regenerates from those archives through a single versioned script that asserts an md5 checksum for every source file it loads, so a changed artifact fails the build rather than silently plotting stale data. Hypotheses, metrics, force-rank controls, and decision bands were registered in design documents before training, and every deviation (one metric re-registration, logged before the affected sweep) is recorded inline in those documents. The registered falsifiers stand open: force-rank-($K-1$) at $\geq 0.9\times$ ceiling at multiple (d, K) points, or a $d \geq 32$ cell clearing the 0.7 bar, would overturn the causal claim. The force-rank staircase aggregate retains one recovery figure per (d, K, k) cell, not per-seed instability diagnostics; the necessity direction (recovery ≈ 0 below K) is robust to the cause of any individual sub-K collapse, while the smoothness of the post-knee approach should be read at the aggregate level only. The analysis pipeline emits benign BLAS `RuntimeWarning`s from near-singular intermediate products on some platforms; every decomposition output is asserted finite before use, so these do not affect any reported number.

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